

$$AB = \begin{bmatrix} 14 & 3 \\ -2 & -6 \end{bmatrix}$$

$$(BA)_{11} = 5 \cdot 2 + 1 \cdot 4 = 14$$

$$BA = \begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix} = \begin{bmatrix} 10 & 2 \\ 29 & -2 \end{bmatrix}$$

$$\left. \begin{array}{l} (BA)_{11} = 2 \cdot 5 + 0 \cdot 3 = 10 \quad (BA)_{12} = 2 \cdot 1 + 0 \cdot (-2) = 2 \\ (BA)_{21} = 4 \cdot 5 + 3 \cdot 3 = 29 \quad (BA)_{22} = 4 \cdot 1 + 3 \cdot (-2) = -2 \end{array} \right\} \text{Row-Column Rule}$$

$$\therefore BA = \begin{bmatrix} 10 & 2 \\ 29 & -2 \end{bmatrix} \quad \underline{AB \neq BA} \quad \square$$

Warnings :

1. In general, $AB \neq BA$.
2. The cancellation laws do not hold for matrix multiplication. That is, if $AB = AC$, that is not true in general that $B = C$. (See Exercise 10).
3. If a product AB is the zero matrix, you cannot conclude in general that either $A = 0$ or $B = 0$. (See Exercise 12)

Powers of a Matrix

If A is an $n \times n$ matrix and if k is a positive integer, then A^k denotes the product of k copies of A :

$$A^k = \underbrace{A \cdots A}_k$$

- If A is nonzero and if x is in $\mathbb{R}^n$, then $A^k x$ is the result of left-multiplying x by A repeatedly k times.
- If $k=0$, then $A^0 x$ should be x itself.
- A^0 is interpreted as the identity matrix.

The Transpose of a Matrix

Given an $m \times n$ matrix A , the transpose of A is the $n \times m$ matrix, denoted by A^T , whose columns are formed from the corresponding rows of A .

Example 8: Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $B = \begin{bmatrix} -5 & 2 \\ 1 & -3 \\ 0 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 1 & 1 \\ -3 & 5 & -2 & 7 \end{bmatrix}$

Then $A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$, $B^T = \begin{bmatrix} -5 & 1 & 0 \\ 2 & -3 & 4 \end{bmatrix}$, $C^T = \begin{bmatrix} 1 & -3 \\ 1 & 5 \\ 1 & -2 \\ 1 & 7 \end{bmatrix} \quad \square$